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Education

PhD, Mathematics, *Dartmouth College*, Hanover, NH Fall 2018 - Spring 2023

Advisor: John Voight

Thesis: *Counting elliptic curves with a cyclic m -isogeny over $\mathbb{Q}$*

MA, Mathematics, *Dartmouth College*, Hanover, NH Fall 2018 - Winter 2020

Advisor: John Voight

MS, Mathematics, *Brigham Young University*, Provo, UT Winter 2017 - Summer 2018

Advisors: Michael Griffin and Paul Jenkins

Thesis: *The arithmetic of modular grids*

BS, Mathematics, *Brigham Young University*, Provo, UT Fall 2013 - Fall 2016

Professional Experience

AI/ML Engineering Manager at ARKA Winter 2025 - Present

My responsibilities are threefold: I innovate in AI and algorithms as an engineer, serve as line manager for both junior and senior AI/ML developers, and act as my company's Algorithm SME. As an engineer, I do cutting-edge work on LLMs, with networks, in ELINT, and more. I also co-write program proposals, serve as a tech lead for a small team, and architect best practices in AI across my program. As a line manager, I nurture a personal rapport with my people, advocate for their interests, and facilitate their work on projects that engage them and enable them to grow in the ways they wish. I am also on the front lines of company recruitment. As Algorithm SME, I liaise with project leads to provide technical support and connect them with the best employees for their problems, and organize and deliver presentations on both foundational and advanced algorithm topics.

Principal AI/ML Engineer at ARKA Summer 2023 - Winter 2025

I spearheaded the development of advanced AI solutions across multiple projects, innovating in areas such as cognitive agents, signal segmentation systems, and satellite scheduling algorithms. I conducted novel research within an Agile framework and served as Scrum Master and AI/Algorithms Lead on one project. I delivered dozens of presentations to company leadership and peers on myriad computational and mathematical topics, from automata to quantum computing.

CCR-Princeton SCAMP Summer Associate Summer 2022

I produced research in the discipline of advanced cryptography. My work included design and vulnerability analysis of novel cryptographic protocols, as well as practical and theoretical work on implementations of homomorphic encryption and secure multi-party computation.

NSA Graduate Mathematics Program Intern

Summer 2021

I produced research focused on distinguishing between human and synthetic audio, using techniques from machine learning, probability theory, and topological data analysis.

Academic Researcher at BYU and Dartmouth

Fall 2013 - Spring 2023

At BYU, I proved results about modular forms and algorithms on graphs. At Dartmouth, I instigated successful research into summing divergent series, the Prisoner's Dilemma, and the Ramanujan-Robin criterion. My thesis develops the arithmetic statistics of elliptic curves with cyclic m -isogeny. See Research Publications and Research Preprints below.

Math Instructor at BYU and Dartmouth

Winter 2017 - Spring 2023

I taught 30-student calculus courses in three different styles: lecturing, virtually with a “flipped classroom”, and with handouts and group exercises. I TAed for lower-division courses, and by special request for upper-division courses as well. See Teaching Experience below.

Technical Skills

Strong with... Python • NumPy • SciPy • scikit-learn • PyTorch • MLflow • Git • Bash/Linux

Experience with... Docker • Lightning • DSPy • Microsoft Agent Framework

Research Publications

A family of analogues to the Robin criterion, with Steve Fan and Mits Kobayashi

The Ramanujan Journal **70**, article 40 (2026). arXiv: 2511.02106

The Robin criterion states that the Riemann hypothesis is equivalent to the inequality $\sigma(n) < e^\gamma n \log \log n$ for all $n > 5040$, where $\sigma(n)$ is the sum of divisors of n , and γ is the Euler–Mascheroni constant. Define $\sigma^{[k]}(n) := \sum_{[d_1, \dots, d_k]=n} d_1 \cdots d_k$, where $[d_1, \dots, d_k]$ is the least common multiple of $d_1, \dots, d_k$. These functions behave asymptotically like $\sigma(n)^k$ as $k \rightarrow \infty$. We prove the following analogue of the Robin criterion: for any $k \geq 2$, the Riemann hypothesis holds if and only if $\sigma^{[k]}(n) < \frac{(e^\gamma n \log \log n)^k}{\zeta(k)}$ for all $n > 2162160$, where ζ is the Riemann zeta function.

Positive spoof Lehmer factorizations, with Guntas Singh

Integers **26**, A33 (2026). arXiv: 2409.17076

We investigate the integer solutions of Diophantine equations related to Lehmer's totient conjecture. We give an algorithm that computes all nontrivial positive spoof Lehmer factorizations with a fixed number of bases r , and enumerate all nontrivial positive spoof Lehmer factorizations with 6 or fewer factors.

A database of basic numerical invariants of Hilbert modular surfaces, with Eran Assaf, Angelica Babei, Ben Breen, Edgar Costa, Juanita Duque-Rosero, Aleksander Horawa, Jean Kiefer, Avinash Kulkarni, Sam Schiavone, and John Voight

Contemp. Math., vol. 796, 2024, Amer. Math. Soc., Providence, RI, 285–312. arXiv: 2301.10302

We describe algorithms for computing geometric invariants for Hilbert modular surfaces, and we report on their implementation.

Counting elliptic curves over the rationals with a 7-isogeny, with John Voight

Research in Number Theory **9**, 75 (2023). arXiv: 2212.11354

We count by height the number of elliptic curves over the rationals, both up to isomorphism over the rationals and over an algebraic closure thereof, that admit a cyclic isogeny of degree 7.

Reactive means in the Iterated Prisoner's Dilemma, with *Caroline Hammond and Feng Fu*
Applied Mathematics and Computation **458** 128201 (2023). arXiv: 2302.13909

The Iterated Prisoner's Dilemma (IPD) is a well studied framework for understanding direct reciprocity and cooperation in pairwise encounters. However, measuring the morality of various IPD strategies is still largely lacking. Here, we partially address this issue by proposing a suit of plausible morality metrics to quantify four aspects of justice. We focus our closed-form calculation on the class of reactive strategies because of their mathematical tractability and expressive power. We define reactive means as a tool for studying how actors in the IPD and Iterated Snowdrift Game (ISG) behave under typical circumstances. We compute reactive means for four functions intended to capture human intuitions about "goodness" and "fair play". Two of these functions are strongly anticorrelated with success in the IPD and ISG, and the other two are weakly anticorrelated with success. Our results will aid in evaluating and comparing powerful IPD strategies based on machine learning algorithms, using simple and intuitive morality metrics.

Seven internal NSA papers, with *IDA Research Team*

Summer 2022.

Abstracts redacted.

9 internal IDA blog posts

Summer 2022.

Abstracts redacted.

Generating and detecting synthetic speech: a technical report, with *NSA Research Team*
Internal NSA paper (Summer 2021).

We present methods for distinguishing between human and synthesized audio, assess their efficacy, and suggest directions for future development. We also present methods for generating realistic-sounding synthetic speech that mimics a human voice, and for confusing machine learning systems such as speaker ID and detection algorithms into believing the generated synthetic speech is real.

The arithmetic of modular grids, with *Michael Griffin and Paul Jenkins*

Mathematika **68**, 1080-1119 (2022). arXiv: 2012.14403

A modular grid is a pair of sequences $(f_m)_m$ and $(g_n)_n$ of weakly holomorphic modular forms such that for almost all m and n , the coefficient of q^n in f_m is the negative of the coefficient of q^m in g_n . Zagier proved this coefficient duality in weights $1/2$ and $3/2$ in the Kohnen plus space, and such grids have appeared for Poincaré series, for modular forms of integral weight, and in many other situations. We give a general proof of coefficient duality for canonical row-reduced bases of spaces of weakly holomorphic modular forms of integral or half-integral weight for every group $\Gamma \subseteq \mathrm{SL}_2(\mathbb{R})$ commensurable with $\mathrm{SL}_2(\mathbb{Z})$. We construct bivariate generating functions that encode these modular forms, and study linear operations on the resulting modular grids.

Odd, spoof perfect factorizations, with *Nickolas Andersen, Spencer Durham, Michael Griffin, Jonathon Hales, Paul Jenkins, Ryan Keck, Hankun Ko, Eric Moss, Pace Nielsen, Kyle Niendorf, Vandy Tombs, Merrill Warnick, and Dongsheng Wu*

Journal of Number Theory **234**, 31-47 (2022). Quanta article. Veritasium video. arXiv: 2006.10697

We investigate the integer solutions of Diophantine equations related to perfect numbers. These solutions generalize the example, found by Descartes in 1638, of an odd, "spoof" perfect factorization $3^2 \cdot 7^2 \cdot 11^2 \cdot 22021^1$. More recently, Voight found the spoof perfect factorization $3^4 \cdot 7^2 \cdot 11^2 \cdot 19^2 \cdot (-127)^1$. No other examples appear in the literature. We compute all nontrivial, odd, primitive spoof perfect factorizations with fewer than seven bases – there are twenty-one in total. We show that the structure of odd, spoof perfect factorizations is extremely rich, and there are multiple infinite families of them. This implies that certain approaches to the odd perfect number problem that use only the multiplicative nature of the sum-of-divisors function are unworkable. On the other hand, we prove that there are only finitely many nontrivial, odd, primitive spoof perfect factorizations with a fixed number of bases.

Zagier duality for level p weakly holomorphic modular forms, with Paul Jenkins

The Ramanujan Journal **50**, 93–109 (2019). arXiv: 1709.10023

We prove Zagier duality between the Fourier coefficients of canonical bases for spaces of weakly holomorphic modular forms of prime level p with $11 \leq p \leq 37$ with poles only at the cusp at ∞ , and special cases of duality for an infinite class of prime levels. We derive generating functions for the bases for genus 1 levels.

Graphs with the strong Havel-Hakimi property, with Michael Barrus

Graphs and Combinatorics **32**, 1689–1697 (2016). arXiv: 1505.00085

The Havel–Hakimi algorithm iteratively reduces the degree sequence of a graph to a list of zeroes. As shown by Favaron, Mahéo, and Sacle, the number of zeroes produced, known as the residue, is a lower bound on the independence number of the graph. We say that a graph has the strong Havel–Hakimi property if in each of its induced subgraphs, deleting any vertex of maximum degree reduces the degree sequence in the same way that the Havel–Hakimi algorithm does. We characterize graphs having this property (which include all threshold and matrogenic graphs) in terms of minimal forbidden induced subgraphs. We further show that for these graphs the residue equals the independence number, and a natural greedy algorithm always produces a maximum independent set.

Books

Puzzle and Proof: A Decade of Problems from the Utah Math Olympiad, with Samuel Dittmer, Hiram Golze, and Caleb Stanford

AK Peters/CRC Recreational Mathematics Series.

This book is a collection of 70 mathematical problems and puzzles from the first ten years of the Utah Math Olympiad (UMO), 2013–2022. These problems are distinguished in two respects. First, they aim to be *understandable* to an advanced high school audience, even if solving them can sometimes be quite difficult. Second, all of the problems ask for not only an answer, but a *proof*.

Research Preprints

The Tropical Algebra of Binary-Tree Height

Preprint. arXiv: 2608.12401

The binary-tree height recursion defines an algebra $\mathcal{H}$ on $\mathbb{N} \cup \{-\infty\}$, with join given by $\max$ and product $a \star b = \max\{a, b\} + 1$. We show that weighted evaluation of a labelled tree depends only on the greatest depth of each label. Single-tree profiles are exactly the vectors satisfying the binary Kraft inequality, while finite joins realize every vector in $(\mathbb{N} \cup \{-\infty\})^n$; hence the n -variable term operations form the free algebra $\mathcal{H}^n$. We also classify $\mathcal{H}$'s compatible semilattice operation, subalgebras, endomorphisms, congruences, and finite quotients, and recover the dyadic-composition spectrum at the full-linear boundary.

The Gregory function and its completed Gregory transform

Preprint. arXiv: 2608.05189

We study the entire interpolation $\mathcal{G}(z) = \int_0^1 \binom{x}{z} dx$ of the Gregory coefficients. Its completion satisfies the positive Markov-transform identity $\frac{\pi z}{\sin(\pi z)} \mathcal{G}(z) = \sum_{n=1}^{\infty} \frac{n|G_n|}{n-z}$. Consequently, every zero is real and simple; the negative zeros are the integers $-1, -2, \dots$, and one zero ρ_n lies in each $(n, n+1)$. We derive complete logarithmic asymptotics for $\rho_n - n$, determine the Cartwright growth and canonical products of $\mathcal{G}$, and realize $1/\rho_n$ spectrally. The resulting relative determinant yields $\gamma = \sum_{n=1}^{\infty} (1/n - 1/\rho_n)$.

A weighted angle distance on strings

Preprint. arXiv: 2604.20633

We define a multi-scale metric d_ρ on strings by aggregating angle distances between all n -gram count vectors with exponential weights ρ^n . We benchmark d_ρ in DBSCAN clustering against edit and n -gram baselines, give a linear-time suffix-tree algorithm for evaluation, prove metric and stability properties, and characterize isometries.

Multiplicative summations into algebraically closed fields, *with Robert Dawson*

Preprint. arXiv: 2111.09938

In this paper, extending our earlier program, we derive maximal canonical extensions for multiplicative summations into algebraically closed fields. We show that there is a well-defined analogue to minimal polynomials for a series algebraic over a ring of series, the “scalar polynomial”. When that ring is the domain of a summation $\mathfrak{S}$, we derive the related concepts of the $\mathfrak{S}$ -minimal polynomial for a series, which is mapped by $\mathfrak{S}$ to a scalar polynomial. When the scalar polynomial for a series has the form $(t - a)^n$, a is the unique value to which the series can be mapped by an extension of the original summation.

Telescopic, multiplicative, and rational extensions of summations, *with Robert Dawson*

Preprint. arXiv: 2105.04592

A summation is a shift-invariant R -module homomorphism from a submodule of R to R or another ring. Dawson formalized a method for extending a summation to a larger domain by telescoping. In this paper, we revisit telescoping, we study multiplicative closures of summations (such as the usual summation on convergent series) that are not themselves multiplicatively closed, and we study rational extensions as a generalization of telescoping.

Expository Notes

Integral-differential calculus

Unpublished. arXiv: 2604.19074

We give an exposition of the Newton-Leibniz calculus. We begin by defining the integral as a limit of Riemann sums, verify the integrals of the standard catalog of functions by direct manipulation, prove the substitution lemmas as theorems about Riemann sums, cross the Fundamental Theorem of Calculus, and harvest the differential calculus on the other side.

On the symmetry of evidential support

Unpublished. arXiv: 2603.29786

For events A and B , we have $\mathbb{P}(A \mid B) > \mathbb{P}(A \mid \neg B)$ if and only if $\mathbb{P}(B \mid A) > \mathbb{P}(B \mid \neg A)$ whenever all four quantities are defined. This note gives seven different proofs of this fact and develops a surrounding web of interpretations.

Minimalist practical numbers

Unpublished.

A natural number n is practical if every smaller number can be written as a sum of distinct divisors of n . We say that a practical number n is minimalist if this representation is unique. In this note, we prove that a practical number is minimalist if and only if it is a power of 2.

Fast-growing series are transcendental, *with Robert Dawson*

Unpublished. arXiv: 2102.12995

Let R be a subring of $\mathbb{C}[[z]]$, and let $X \in \mathbb{C}[[z]]$. The Newton-Puiseux Theorem implies that if the coefficients of X grow sufficiently rapidly relative to the coefficients of the series in R , then X is transcendental over R . We provide an alternative proof of this result by establishing a relationship between the coefficients of $A(X)$ and $A'(X)$, where $A(T)$ is a polynomial over $\mathbb{C}[[z]]$.

Teaching Experience

Dartmouth College, Hanover, NH (Instructor)

Math 8 (Calculus of One and Several Variables)

Fall 2021

Math 1 (Algebra and Calculus)

Fall 2020

Dartmouth College, Hanover, NH (Teacher's Assistant)

Math 100 / Computer Science 49/149 (Random Walk)

Spring 2022

Math 100 / Computer Science 49/149 (Decision Theory)

Spring 2021

Math 23 (Differential Equations)

Winter 2020

Math 3 (Calculus)	Fall 2019
Math 22 (Linear Algebra)	Spring 2019
Math 22 (Linear Algebra)	Fall 2018
Brigham Young University , Provo, UT (Instructor)	
Math 112 (Calculus I)	Summer 2017
Brigham Young University , Provo, UT (Teacher's Assistant)	
Math 112 (Calculus I)	Fall 2017
Math 113 (Calculus II)	Winter 2017
Brigham Young University , Provo, UT (Grader)	
Math 570 (Matrix Analysis)	Winter 2017
Math 112 (Calculus I)	Fall 2015

Awards and Achievements

Certifications

PSM I Certification

Awards and Fellowships

Dartmouth Graduate Fellowship	Fall 2018 - Summer 2023
Gridley Fund for Graduate Mathematics	Fall 2018 - Summer 2019
NSF Graduate Research Fellowship Honorable Mention	April 2018
BYU Academic Scholarship	Spring 2015 - Winter 2016
BYU Math Department Scholarship (Putnam)	Fall 2016
BYU Math Department Scholarship (Putnam)	Fall 2015
BYU Academic Scholarship	Fall 2013 - Winter 2014

Achievements and Honors

3 rd Place in the Virginia Tech Regional Math Competition	October 2016
2 nd Place in the Search for High Rank Elliptic Curve at 2016 Connecticut Summer School in Number Theory	August 2016
Gold Palm Eagle Scout	December 2012
Black Belt (Taekwondo)	August 2012

Special Training

Dale Carnegie Training: Leadership Training for Managers	Summer 2025
Dartmouth Mathematics Teaching Seminar	Spring 2020
BYU Mathematics Teaching Seminar	Fall 2018

Leadership and Community Service

Referee

Experimental Mathematics • Integers • Journal of Mathematical Analysis and Applications • Journal of Number Theory • Mathematische Zeitschrift • Theory and Decision

Reviewer

zbMATH Open • CRC Press

K-12 Outreach

Utah Math Olympiad Committee Member	2015 - Present
Lumiere Mentor	2024
Activity Leader at Dartmouth Science Day (Games & Surreal Numbers)	April 8, 2023
Activity Leader at Lebanon Ward Pi Day (Trachtenberg Arithmetic & Random Walk)	March 25, 2023
Session Leader at Dartmouth Sonia Kovalevsky Day (Trachtenberg Arithmetic)	May 21, 2022
Activity Leader at Lebanon Ward Pi Day (Random Walk)	April 8, 2022
Exploring Mathematics Camp Leader (Graph Theory)	July 27-31, 2020
Exploring Mathematics Camp Leader (Cryptography)	July 13-17, 2020
Activity Leader at Lebanon Ward Pi Day (Random Walk)	March 7, 2020
Volunteer at Dartmouth Sonia Kovalevsky Day	May 11, 2019
Activity Leader at Dartmouth Science Day (Nim)	May 4, 2019
Math Circles Guest Speaker (Tropical Algebra)	February 24, 2018
Math Circles Guest Speaker (Continued Fractions)	October 21, 2017
Proofreader for Utah State Math Contest	February 2017
Math Circles Counselor	2013 - 2015

Dartmouth Graduate Student Council

Ad Hoc Healthcare Committee Founder and Co-Chair	2022 - 2023
◦ <i>Budget Committee Member</i>	Summer 2022
Finance Officer	2021 - 2022
◦ <i>Budget Committee Member</i>	Summer 2021
Representative for Math Department	2020 - 2021
◦ <i>Service Committee Member</i>	2020 - 2021
◦ <i>Budget Committee Member</i>	Summer 2020

Dartmouth Directed Reading Program

Mentor (Fuzzy Logic)	Winter 2023
Mentor (Decision Theory)	Spring 2022
Mentor (Fractional Calculus)	Winter 2022

Other

LMFDB Contributor	2019 - 2023
Dartmouth Algebra and Number Theory Seminar Organizer	2019 - 2022
CCR-P SCAMP Number Theory Discussion Group Organizer	Summer 2022
BYU Putnam Team Captain	2014 - 2016

Thesis Defenses

Counting elliptic curves with a cyclic m-isogeny over $\mathbb{Q}$	April 24, 2023
Dartmouth College PhD Thesis Defense	
The arithmetic of modular grids	June 22, 2018
BYU Master's Thesis Defense	

Invited Talks

Quantum communication & computing 2025 ARKA Tech Day — invited plenary presentation at company R&D strategy retreat	January 27, 2025
Counting elliptic curves with a cyclic m-isogeny Job Talk at Metron	February 9, 2023
A family of analogues to the Ramanujan-Robin criterion Job Talk at Center for Computer Science, Bowie	February 2, 2023
Counting elliptic curves with a 7-isogeny BYU Colloquium	January 19, 2023
Counting elliptic curves with a 7-isogeny Simons Collaboration Annual Meeting	January 11, 2023
Counting 7-isogenies 2023 Joint Mathematics Meetings	January 6, 2023
Counting elliptic curves with a 7-isogeny NSA Seminar (Virtual)	December 15, 2022
Title redacted Next Generation Cryptography at CCR-La Jolla	November 16, 2022
A family of analogues to the Ramanujan-Robin Criterion BYU Number Theory Seminar	October 27, 2022
Title redacted Briefing to General Paul Nakasone, then-Director of the NSA	July 26, 2021
Title redacted Briefing to Center for Computing Sciences, Bowie	June 25, 2021

Other Conference Talks

A family of analogues to the Robin criterion 2026 Joint Mathematics Meetings	January 6, 2026
Intersecting varieties with transcendental graphs Arizona Winter School 2023: Unlikely Intersections	March 8, 2023
The LCM product and Grönwall's theorem 2021 Maine-Québec Number Theory Conference (virtual)	October 3, 2021
Formal summation of divergent series 2020 Conférence de Théorie des Nombres Québec-Maine (virtual)	September 26, 2020
The arithmetic of modular grids 2019 Maine-Québec Number Theory Conference	October 5, 2019
The arithmetic of modular grids 33 rd Automorphic Forms Workshop	March 8, 2019
The arithmetic of modular grids Building Bridges: 4th EU/US Summer School + Workshop on Automorphic Forms	July 17, 2018
Zagier duality in level p modular spaces 32 nd Automorphic Forms Workshop	March 21, 2018

Zagier duality in level p modular spaces 2018 BYU Student Research Conference	March 3, 2018
Zagier duality in level p modular spaces Modular Forms are Everywhere Conference	May 24, 2017
Weakly holomorphic modular forms of level 11 31 st Automorphic Forms Workshop	March 7, 2017
Weakly holomorphic modular forms of level 11 2017 BYU Student Research Conference	March 4, 2017
Congruence relations in modular forms of prime levels greater than 7 2015 BYU Student Research Conference	March 21, 2015
Residues and independence numbers of graphs 2014 BYU Student Research Conference	March 15, 2014

Selected Seminar Talks

Attention is all you need ARKA Paper Book Club	March 27, 2026
From last year to next ARKA Algorithm Development Tech Titan Presentation	February 20, 2026
MLOps ARKA Paper Book Club	January 13, 2026
Distances on probabilities ARKA Artificial Intelligence Tech Titan Presentation	December 12, 2025
Are small language models the future of agentic AI? ARKA Paper Book Club	November 18, 2025
Is chain-of-thought reasoning a mirage? ARKA Paper Book Club	October 28, 2025
Algorithms in 2025: a CCCG/WADS retrospective ARKA Algorithm Development Tech Titan Presentation	September 12, 2025
Bayesian stochastic blockmodeling ARKA Lunch & Learn Series	July 9, 2025
Testing, testing, testing ARKA Algorithm Development Tech Titan Presentation	June 6, 2025
Procrustes analysis ARKA Lunch & Learn Series	April 9, 2025
Algorithms and computational complexity ARKA Algorithm Development Tech Titan Presentation	February 21, 2025
Robust optimization ARKA Algorithm Development Tech Titan Presentation	December 5, 2024
Linear programming ARKA Algorithm Development Tech Titan Presentation	November 21, 2024
The comedy of errors ARKA Algorithm Development Tech Titan Presentation	October 24, 2024

Applied category theory ARKA Algorithm Development Tech Titan Presentation	September 12, 2024
Neurofuzziness ARKA Artificial Intelligence Tech Titan Presentation	June 27, 2024
Fuzzy logic ARKA Algorithm Development Tech Titan Presentation	June 6, 2024
Finite automata for the software engineer ARKA Algorithm Development Tech Titan Presentation	April 25, 2024
Quantum computing IV: Shor's algorithm ARKA Algorithm Development Tech Titan Presentation	March 28, 2024
Quantum computing III: QML ARKA Artificial Intelligence Tech Titan Presentation	March 21, 2024
Spacepower: doctrine for Space Forces ARKA SDA & Space Control Tech Titan Presentation	February 29, 2024
Quantum computing II: algorithm fundamentals ARKA Algorithm Development Tech Titan Presentation	February 29, 2024
Quantum computing I: bits and qubits ARKA Algorithm Development Tech Titan Presentation	February 15, 2024
Neural differential equations ARKA Artificial Intelligence Tech Titan Presentation	January 25, 2024
Topological data analysis ARKA Algorithm Development Tech Titan Presentation	December 7, 2023
Geometric deep learning ARKA Artificial Intelligence Tech Titan Presentation	October 19, 2023
Symbolic segmentation II ARKA Algorithm Development Tech Titan Presentation	September 14, 2023
Symbolic segmentation I ARKA Algorithm Development Tech Titan Presentation	September 14, 2023
The Fox H-function Dartmouth Graduate Student Seminar	March 28, 2023
Geometric deep learning Dartmouth Graduate Student Seminar	February 14, 2023
Nearing nearrings Dartmouth Graduate Student Seminar	September 14, 2022
Universal algebra and coalgebra Dartmouth Graduate Student Seminar	March 30, 2022
Cogalois theory Dartmouth Graduate Student Seminar	February 16, 2022
Cyclic resolution of singularities Reading Seminar on Hilbert Modular Surfaces	January 25, 2022
What if the Riemann hypothesis is false? Dartmouth Graduate Student Seminar	January 19, 2022

Inverse semigroups: groups without identity Dartmouth Graduate Student Seminar	November 9, 2021
Counting 7-isogenies Dartmouth Algebra and Number Theory Seminar	November 8, 2021
A sober look at pointless topology Dartmouth Graduate Student Seminar	September 21, 2021
Fast-growing series are transcendental Dartmouth Graduate Student Seminar (virtual)	April 21, 2021
Coalgebras and Hopf algebras Reading Seminar on Affine Group Schemes (virtual)	April 15, 2021
Absurd equalities and Runge's method: the degenerate case Dartmouth Algebra and Number Theory Seminar (virtual)	March 30, 2021
p-adic Hodge theory Reading Seminar on Classical and Quadratic Chabauty (virtual)	March 2, 2021
Odd, spoof quasiperfect factorizations Dartmouth Algebra and Number Theory Seminar (virtual)	February 9, 2021
A primer in social choice theory Dartmouth Graduate Student Seminar (virtual)	January 20, 2021
Examples of Kedlaya's algorithm Reading Seminar on Classical and Quadratic Chabauty (virtual)	November 19, 2020
The LCM product and Grönwall's theorem Dartmouth Algebra and Number Theory Seminar (virtual)	November 17, 2020
Reactive means and the prisoner's dilemma Dartmouth Graduate Student Seminar (virtual)	October 7, 2020
Formal summation of divergent series: an algebraic approach Dartmouth Algebra and Number Theory Seminar (virtual)	April 28, 2020
On the infinitude of the natural numbers Dartmouth Graduate Student Seminar	February 12, 2020
Variations of Hodge structures Reading Seminar on Shimura Varieties	November 26, 2019
Hodge structures Reading Seminar on Shimura Varieties	November 14, 2019
Savage's expected utility and making good decisions Dartmouth Graduate Student Seminar	November 13, 2019
Geometric and generalized calculus Dartmouth Graduate Student Seminar	October 8, 2019
Summing divergent series Dartmouth Graduate Student Seminar	July 31, 2019
Real analysis: a nonstandard approach Dartmouth Graduate Student Seminar	April 17, 2019
Why save the universe? Set theory with a universal set Dartmouth Graduate Student Seminar	January 16, 2019

The arithmetic of modular grids Dartmouth Graduate Student Seminar	September 26, 2018
Dioids and idempotent geometry II BYU Algebra Seminar	March 13, 2018
Dioids and idempotent geometry I BYU Algebra Seminar	February 27, 2018
Zagier duality in level p modular spaces BYU Number Theory Seminar	February 22, 2018
Average values of arithmetic functions BYU Number Theory Seminar	October 12, 2017
Weakly holomorphic modular forms of level 11 BYU Number Theory Seminar	February 9, 2017
Generating functions for canonical bases of certain level 11 weakly holomorphic modular forms BYU Number Theory Seminar	January 26, 2017
High rank elliptic curves with prescribed torsion BYU Number Theory Seminar	December 1, 2016
Heuristics for elliptic curves of high rank BYU Number Theory Seminar	October 20, 2016
A lemma regarding the Feit-Thompson conjecture BYU Number Theory Seminar	November 12, 2015
Algebraic number theory and the Feit-Thompson conjecture BYU Number Theory Seminar	October 1, 2015
Congruence relations in modular forms of prime levels greater than 7 BYU Number Theory Seminar	March 19, 2015
Hensel's lemma BYU Number Theory Seminar	February 12, 2015
Newton polynomials BYU Number Theory Seminar	March 27, 2014

Conferences and Workshops Attended

2026 <i>Joint Mathematics Meetings</i> Walter E. Washington Convention Center, Washington DC	January 4-7, 2026
CCCG/WADS 2025 York University, Toronto	August 11 - 15, 2025
<i>Qubits 2025: Quantum Realized</i> Virtual (Scottsdale, Arizona)	March 31 - April 1, 2025
2025 <i>Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation Annual Meeting</i> Virtual (Simons Foundation's Gerald D. Fischbach Auditorium, New York City)	January 15-16, 2025

2024 <i>Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation Annual Meeting</i> Virtual (Simons Foundation's Gerald D. Fischbach Auditorium, New York City)	January 11-12, 2024
<i>Big Data Conference 2023</i> Virtual (Harvard University, Cambridge)	August 31 - September 1, 2023
<i>Hilbert Modular Forms Infrastructure Week 4</i> Dartmouth College, Hanover	May 8-12, 2023
<i>Arizona Winter School 2023: Unlikely Intersections</i> University of Arizona, Tucson	March 4-8, 2023
2023 <i>Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation Annual Meeting</i> Simons Foundation's Gerald D. Fischbach Auditorium, New York City	January 11-12, 2023
2023 <i>Joint Mathematics Meetings</i> Hynes Convention Center, Boston	January 4-7, 2023
<i>Hilbert Modular Forms Infrastructure Week 3</i> Dartmouth College, Hanover	December 12-16, 2022
<i>Next-Generation Cryptography</i> Center for Communications Research, La Jolla	November 14-18, 2022
<i>Modular Curves Workshop 2</i> MIT, Boston	November 5-9, 2022
<i>Big Data Conference 2022</i> Virtual (Harvard University, Cambridge)	August 26, 2022
<i>Graph Fest 2022</i> NSA Internal Conference	June 29-30, 2022
<i>Explicit Methods for Modularity</i> Virtual (Simons Collaboration)	April 11-15, 2022
<i>Arizona Winter School 2022: Automorphic Forms Beyond GL_2</i> Virtual (University of Arizona, Tucson)	March 5-9, 2022
<i>Hilbert Modular Forms Infrastructure Week 2</i> Dartmouth College, Hanover	February 21-25, 2022
2021 <i>Maine-Québec Number Theory Conference</i> Virtual (University of Maine, Orono)	October 2-3, 2021
<i>MathFest 2021</i> NSA Internal Conference	June 4, 2021
<i>Spring 2021 Algebraic Geometry Northeastern Series</i> Virtual (Brown University, Providence)	May 4-5, 2021
21 st <i>Algebra, Geometry and Combinatorics Day</i> Virtual (University of Notre Dame, Notre Dame)	April 10, 2021
2021 <i>Joint Mathematics Meetings</i> Virtual (American Math Society, Mathematical Association of America)	January 6-9, 2021

<i>Fall 2020 Algebraic Geometry Northeastern Series</i> Virtual (Stony Brook University, Stony Brook)	October 23-25, 2020
2020 <i>Conférence de Théorie des Nombres Québec-Maine</i> Virtual (Université Laval, Québec)	September 26-27, 2020
14 th <i>Algorithmic Number Theory Symposium</i> Virtual (University of Auckland, Auckland)	June 29 - July 4, 2020
<i>UNCG Summer School in Computational Number Theory and Algebra: Ergodic Theory with Applications to Continued Fractions</i> Virtual (University of North Carolina, Greensboro)	May 18-22, 2020
2019 <i>Maine-Québec Number Theory Conference</i> University of Maine, Orono	October 5-6, 2019
<i>Fall 2019 Algebraic Geometry Northeastern Series</i> Boston College, Boston	September 20-22, 2019
33 rd <i>Automorphic Forms Workshop</i> Duquesne University, Pittsburgh	March 6-10, 2019
<i>Building Bridges: 4th EU/US Summer School + Workshop on Automorphic Forms and Related Topics</i> Alfréd Rényi Institute of Mathematics, Budapest	July 9-20, 2018
32 nd <i>Automorphic Forms Workshop</i> Tufts University, Medford	March 19-22, 2018
<i>Modular Forms are Everywhere</i> Max Planck Institute for Mathematics, Bonn	May 15-26, 2017
31 st <i>Automorphic Forms Workshop</i> East Tennessee State University, Johnson City	March 6-9, 2017
2017 <i>BYU Student Research Conference</i> Brigham Young University, Provo	March 4, 2017
2016 <i>Connecticut Summer School in Number Theory</i> University of Connecticut, Storrs	August 8-14, 2016
2015 <i>BYU Student Research Conference</i> Brigham Young University, Provo	March 21, 2015
2014 <i>BYU Student Research Conference</i> Brigham Young University, Provo	March 15, 2014